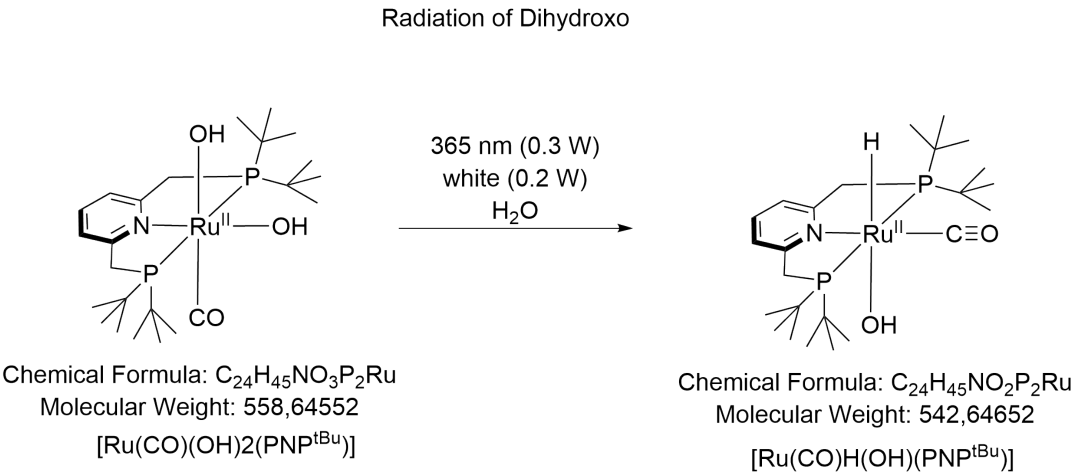
AE-426: Radiation of [Ru(CO)(OH)2(PNP)] in water with liquid phase O2 measurment

Date: 2025-01-23
Tags: Radiation O2 PNP [Ru(CO)(OH)2(PNP)] AE liquidphase
Category: Two-photon
Status: Done
Created by: Alexander Eith

Reaction scheme/sample structure



Literature/reference experiments

Literature	https://doi.org/10.1039/D1EE01053K
Reproduction	/
Related experiments	Two-photon - AE-418: Radiation of [Ru(CO)(OH)2(PNP)] in water with liquid phase O2 measurment Two-photon - AE-419: Radiation of water with liquid phase O2 measurment, analog to AE-418

Reagents

Name	CAS/Experiment number	Amount [µmol]	Equivalents	Mass$_{theo}$ [mg]	Mass$_{exp}$ [mg]	Molar mass [g/mol]	Volume [ml]
[Ru(CO)(OH)2(PNP{tBu})]	Two-photon - AE-382: Synthesis of [Ru(CO)(OH)2(PNPtBu)] using Ag2O, 5 h (**AE-382-3**)	/	/	/	2.31	558.65	/
Milli-Q water	GD-018: Redegassing of H2O	/	/	/	/	/	0.92

Irradiation Parameters

Power measured using [Power Meter] 843-R-USB + 919P-020-12 unless specified otherwise.

	Light Source Name	Wavelength [nm]	Power Setting [mW]
First light source	Light Source - LCS-0365-76-22	365 nm	300
Second light source	Light Source - LCS-6500-65-22	white	200

Used beam combiner [Name or None]	Beam Combiner - LCS-BC25-0409
Irradiation distance [cm]	9
Thermostat temperature [°C]	25
Stirring speed [rpm]	1000
Irradiation start [s]	2880
Irradiation stop [s]	8720

Procedure/observations

All steps were carried out using [Protocol] Schlenk Technique. All steps were carried out in the dark, unless other specified.

Date	Time	Step	Observations
23.01		A 10 mL screw cap schlenk flask was prepared	
	13:35	To the flask [Ru] was added under air	
		The flask was set under Ar	
	13:55	To the flask water (0.92 mL) was added	
		The mixture was stirred	brwonish suspension in water.jpg
	14:00	The flask stored in Equipment - Freezer Lab E004 CEEC II under Ar in the dark	

24.01		A screw cap 5 mL irradiation flask was prepared	
	9:30	The 10 mL flask was taken out of the freezer	
		The power of the LEDs 365 nm: 300 mW and white: 200 mW were adjusted according to [Protocol] Irradiation setup	power setup.jpg
	9:40	The 10 mL flask was stirred at 600 rpm	after stir.jpg no change
	10:00	The suspension was filtered into the 5 mL flask using a PA syringe filter (pore diameter = 0.2 µm	yellowish solution after filt.jpg
	10:04	The Robust oxygen probe was attached to the flask using a bola fitting (diameter = 3 mm)	
	10:05	The flask was moved to the setup	in setup.jpg
	10:07	The logging was started (AE-426-1-Ch2)	
	10:08	The flask was flushed with argon for 130 s (strong Ar flow in gas phase)	
	10:11 - 10:35	The Schlenk was degassed according to Protocol - Degassing and O2 Measurement with a sensor cap with air-stable substances in the Photochem. setup	c(O2) = approx. -0.6 µmol/L during degas.jpg no change
	10:35	The degassing stopped by removing the PTFE cannula according to Protocol - Degassing and O2 Measurement with sensor cap with air stable substances	
	10:37	The Log (-1-Ch2) was stopped	c(O2) = approx. 0.2 µM AE-426-1.png AE-426-1-Ch2.txt
	10:37	The logging was started (AE-426-2-Ch2)	
	10:37 - 11:25	The sample was allowed to equillibrate according to Protocol - Degassing and O2 Measurement with sensor cap with air stable substances	Equilibration duration: 2880 s
	11:25	The light sources were turned on	
	13:03	The light source was turned off	Irradiation duration: 5840 s

	13:03 - 15:13	The sample was allowed to equilibrate according to Protocol - Degassing and O2 Measurement with sensor cap with air stable substances	Equilibration duration: 7834 s after irrad.jpg orange solution with small amount of solids
	13:54	The logging was stopped (-2-Ch2)	AE-426-2-Ch2.txt AE-426-2.png

Analysis

Date	Time	Sample name	Analysis method	Analytic device	Solvent	Raw Data	Processed Data	Comparative Data	Interpretation
24.01	10:07	AE-426-1-Ch2	Optical O2 detection	Equipment - Firesting Fiber-Opitc Oxygen Meter 2 Channel	H2O	AE-426-1-Ch2.txt	AE-426-1.png	Two-photon - AE-402: Radiation of [Ru(CO)(OH)2(PNP)] in water with liquid phase O2 measurment (**AE-402-1-Ch2** and **AE-402-2-Ch2**)	Degassing worked. After stop increase to 0.2 µM
24.01	10:37	AE-426-2-Ch2	Optical O2 detection	Equipment - Firesting Fiber-Opitc Oxygen Meter 2 Channel	H2O	AE-426-2-Ch2.txt	AE-426-2.png	Two-photon - AE-402: Radiation of [Ru(CO)(OH)2(PNP)] in water with liquid phase O2 measurment (**AE-402-3-Ch2**) Two-photon - AE-418: Radiation of [Ru(CO)(OH)2(PNP)] in water with liquid phase O2 measurment (**AE-418-2-Ch2**)	Very similar to AE-402 and AE-418. Slow decrease during equilibration. Sharp drop to -0.6 at start of irrad, than graduale increase by 0.1 over approx. 5 min, than slow decrease similar to decrease before irrad. At stop of irrad drop to -0.05 µmol/L, than slow increase over time (maybe slowly flattining)

Result

Calibration using static degassed conditions did not help to get rid of the artefacts. Otherwise reproducible

Linked experiments

- AE-236: Irradiation of PhPDA (AE-234), 1.5 mg/mg SDS, 2 mg/mL PhPDA

- AE-237: Irradiation of PhPDA (AE-234), 1.5 mg/mg SDS, 2 mg/mL PhPDA I

- AE-245: Synthesis of poly(1,4-diphenylbutadiyne)

- AE-249: Irradiation of PhPDA (AE-245), 1.5 mg/mg SDS, 2 mg/mL PhPDA

- AE-254: Irradiation of PhPDA (AE-245), 1.5 mg/mg SDS, 2 mg/mL PhPDA reproduction

- AE-257: Synthesis of poly(1,4-diphenylbutadiyne)

- AE-258: Irradiation of PhPDA (AE-257), 1.5 mg/mg SDS, 2 mg/mL PhPDA

- AE-259: Irradiation of PhPDA (AE-257), 1.5 mg/mg SDS, 2 mg/mL PhPDA after 1 d

- AE-260: Irradiation of PhPDA (AE-257), 1.5 mg/mg SDS, 2 mg/mL PhPDA after 2 d

- AE-262: Irradiation of PhPDA (AE-257), 1.5 mg/mg SDS, 2 mg/mL PhPDA after 3 d

- AE-276: Synthesis of poly(1,4-diphenylbutadiyne), 1.2 g scale

- AE-279: Irradiation of PhPDA (AE-276-2), 1.5 mg/mg SDS, 2 mg/mL PhPDA

- AE-282: Irradiation of PhPDA (AE-276-1), 1.5 mg/mg SDS, 2 mg/mL PhPDA

- GD-018: Redegassing of H2O

- AE-313: Synthesis of PPy via Fenton reaction with FeSO4 and 10 eq. H2O2 I

- AE-332: Irradiation of PPy (AE-312), 2 mg/mL

- AE-333: Irradiation of PPy (AE-313), 2 mg/mL

C3N4 - AE-398: Irradiation of C3N4-Co-0.2 from Lingli Ni

C3N4 - AE-399: Irradiation of C3N4-Co-0.4 from Lingli Ni I

Two-photon - AE-358: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-310) 365 nm and white, 50 mM KOH

Two-photon - AE-382: Synthesis of [Ru(CO)(OH)2(PNPtBu)] using Ag2O, 5 h

Two-photon - AE-386: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-382) 365 nm and white, 50 mM KOH in D2O I

Two-photon - AE-402: Radiation of [Ru(CO)(OH)2(PNP)] in water with liquid phase O2 measurment

Two-photon - AE-418: Radiation of [Ru(CO)(OH)2(PNP)] in water with liquid phase O2 measurment

Two-photon - AE-419: Radiation of water with liquid phase O2 measurment, analog to AE-418

Linked resources

Archived - Degassing procedure for photocatalytic water splitting by polymeric photocatalysts

Beam Combiner - Beam Combiner LCS-BC25-0409 -1

Beam Combiner - Beam Combiner LCS-BC25-0409 -2

Equipment - Firesting Fiber-Optic Oxygen Meter 2 Channel (Firesting 1)

Equipment - Freezer Lab E004 CEEC II

Light Source - UHP LED 405 nm

Light Source - UHP LED 6500 K (white)-1

Light Source - HP LED 405 -2

Light Source - HP LED 6500 K (white) -2

Protocol - Irradiation setup

Protocol - UV vis measurements (BioTek PowerWave HT)

Protocol - Determination of H2O2 by Spectroquant

Protocol - Degassing and O2 Measurement with sensor cap with air stable substances

Protocol - Determination of H2O2 in the suspensions of photocatalysts using H2O2 test strips

Protocol - Ultrasonication with the UP 200st from hilscher in ZAF 208

Protocol - H2O2 photometric detection with DPD-POD method for irradiated suspensions of photocatalysts

Attached files

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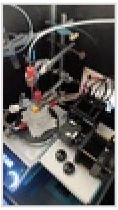

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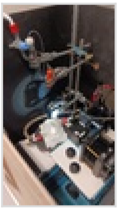

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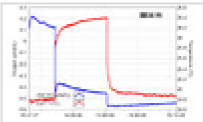

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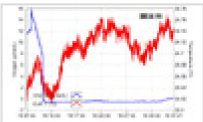

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